

Instawards Statement of Work (SOW)
30-Day Scoped Engagement

1. Project & Team Information

Project Name:	INIT.AI
Builder / Team Name:	RAIN Studio/Systems
Primary Contact (Name + Email):	Rain Regencia + angelorainregencia@gmail.com
Team Members:	Angelo Rain Regencia (Project Manager) Sean Astin Navarro (Full-stack Developer) Marc Joshua Valenzuela (Quality Assurance)
Ambassador Chapter:	Stellar Philippines
Ambassador Chapter Lead:	Armielyn Obinguar
Date Submitted:	August 17, 2026
Suggested Sprint Start Date:	Upon approval <i>(within SDF's 3-5 business-day review window)</i>

2. Instawards Overview & Intent

2.1 Instawards Purpose (for Builder Context)

Instawards are designed to support short, clearly scoped, execution-focused work that helps a project make tangible progress toward building on Stellar. Instawards are meant to fund specific, achievable outcomes that can be completed and demonstrated within 30 days or less.

This SOW represents a shared commitment between the Builder and the Ambassador Chapter Lead on what will be delivered, why it matters, and how success will be verified.

3. Problem Statement & Objective

Instawards Statement of Work (SOW)
30-Day Scoped Engagement

Problem Being Addressed	What specific problem, gap, or blocker is this Instaward intended to solve?	INIT.AI currently operates as a Web2 geospatial AI platform mapping urban heat islands. However, municipal heat metrics and LGU climate mitigation data lack an immutable, publicly verifiable audit trail on-chain. Without transparent, tamper-proof verification, local government units (LGUs), climate auditors, and donor organizations cannot verify heat reduction progress or safely automate climate micro-grant routing.
Objective of This Instaward	In one or two sentences, what will be true at the end of 30 days if this Instaward is successful?	At the end of 30 days, INIT.AI will have a live, working Web3 verification layer on Stellar Testnet. cryptographically hashed spatial heat attestations via a Rust-based Soroban smart contract, sign transactions using StellarWalletsKit (Freighter) from the INIT.AI dashboard, and audit transaction receipts on Stellar Expert.
<i>Example prompts for builders: What is currently preventing progress? What is unclear, missing, or unbuilt today? Why is this problem worth solving now?</i>		

3.1 Key Outcomes

At the end of this sprint, a tester, climate auditor, or INIT.AI user can:

- Submit a spatial heat report from the INIT.AI dashboard and have it converted into a SHA-256 cryptographic digest by the backend hashing pipeline using the published canonicalization specification.
- Review the proposed attestation, then sign and submit the on-chain transaction (log_heat_index) directly from their own Stellar wallet via StellarWalletsKit (Freighter); the backend only prepares the unsigned XDR and relays the signed XDR and never custodies a private key.

Instawards Statement of Work (SOW)

30-Day Scoped Engagement

- Have a write-once attestation record containing the SHA-256 digest and minimal, non-sensitive metadata recorded in the SpatialAttestationRegistry Soroban contract on Stellar Testnet; the underlying GeoJSON heat report remains off-chain.
- Independently verify an attestation's existence, digest, signing wallet, and confirmation status on Stellar Expert, without treating INIT.AI's database as the source of truth for the on-chain record.

No manual reconciliation or backend override is required for a confirmed on-chain record.

This sprint does not rebuild the INIT.AI mapping platform; it adds one new layer to a system that already produces urban heat-island data today: an on-chain, cryptographically verifiable attestation trail. The chain proves that a particular digest was recorded at a particular time by a signing wallet. It does not prove scientific accuracy, data completeness, LGU authorization, signer identity, or that the underlying heat report is correct or immutable.

Expected ecosystem impact. Adding a verifiable attestation layer to an existing LGU-facing climate platform converts selected heat reports into measurable on-chain Stellar activity while keeping raw spatial data off-chain and reducing the need to trust INIT.AI's database for historical attestation verification.

3.2 The Gap & Current Challenges

Municipal and NGO climate reporting today is dominated by spreadsheets, PDFs, and closed dashboards, simple to edit after the fact and impossible for a donor or auditor to independently verify without trusting the reporting organization outright. INIT.AI's own data today carries the same limitation: it is accurate, but not independently checkable. The ways this kind of data is handled today all fail for open, tamper-evident public-sector verification:

Approach	Key Limitation
Spreadsheet / PDF climate reporting (status quo)	Editable after submission, no tamper-evidence, no independent verification path for an outside auditor.
Centralized SaaS climate dashboards	Data lives in a private database; an auditor must trust the vendor's backend rather than a public record.

Instawards Statement of Work (SOW)
30-Day Scoped Engagement

General-purpose blockchain timestamping services	Confirm that a hash existed at a point in time, but are not purpose-built for spatial heat data, LGU identifiers, or LGU-specific audit workflows.
INIT.AI (with this Instaward)	Purpose-built SHA-256 attestation of spatial heat reports, opaque report reference, version identifier, and timestamp, written immutably to a Soroban contract and independently verifiable on Stellar Expert.

3.3 Why Stellar & Ecosystem Impact

Stellar makes frequent, low-value civic attestations economically viable and independently verifiable. INIT.AI's verification layer is built natively on Stellar developer tooling:

- Stellar technologies used: a Rust + soroban-sdk smart contract (SpatialAttestationRegistry), stellar-cli for WASM build/deployment, StellarWalletsKit (Freighter) for client-side signing, @stellar/stellar-sdk for XDR/transaction handling, and Stellar Testnet Soroban RPC for transaction simulation, submission, and confirmation monitoring.
- Low Testnet transaction cost: the proof of concept is designed around lightweight attestations so recurring civic records remain economically practical; no mainnet cost claim is made in this sprint.
- Fast confirmation target: the dashboard should surface Testnet confirmation within seconds when RPC and network conditions permit, but the UI must not imply finality before Stellar Testnet confirms successful execution.
- On-chain growth: each successfully confirmed heat-report attestation becomes real Testnet activity—a signed transaction and Soroban contract invocation—tied to a public-sector, non-financial use case.

Why now. Municipal and NGO climate reporting is under growing pressure to show verifiable, audit-ready data, as donor funding and LGU climate-mitigation grants increasingly expect proof of measurement rather than self-reported figures. It is far cheaper to establish a lightweight, wallet-signed attestation convention now, while INIT.AI's LGU pilots are still small, than to

Instawards Statement of Work (SOW)

30-Day Scoped Engagement

retrofit trust into a reporting pipeline after donor-facing dashboards are already built on unverifiable data.

3.4 Existing Approaches

Other blockchain-based environmental verification efforts exist, but none combine spatial, LGU-scoped heat attestation with wallet-signed, per-report proof on Stellar:

Platform	Primary Focus
Regen Network	Carbon-credit and ecological-asset MRV on its own Cosmos-based chain (Regen Ledger) - built for credit issuance and trading, not municipal heat-index reporting, and not on Stellar.
Toucan Protocol	Bridges existing carbon registries (e.g., Verra-issued credits) onto Polygon as tokenized assets - a bridging/tokenization layer for credits already verified elsewhere, not a native spatial-attestation contract.
Open Forest Protocol	Blockchain-based MRV focused on forestry and reforestation projects - not urban heat data or LGU civic reporting.
INIT.AI	Purpose-built SHA-256 attestation of urban heat-index reports tied to LGU IDs, written directly to a Soroban contract on Stellar and signed client-side by the reporting party's own wallet.

3.5 Validation Scope & Practical Use Cases

Validation Scope. This Instaward validates on-chain attestation writing (a user-controlled wallet → a write-once SpatialAttestationRegistry entry), wallet-based signing and verification UX, canonical hashing, transaction simulation, confirmation/error handling, and the hashing/RPC

Instawards Statement of Work (SOW)

30-Day Scoped Engagement

bridge that connects INIT.AI's existing spatial data pipeline to Stellar. It does not evaluate the underlying accuracy of INIT.AI's heat-index modeling itself; on-chain attestation proves that a specific digest was signed and recorded at a specific time by a wallet.

Use Cases:

- LGU climate reporting proof of concept: a wallet holder submits a quarterly heat-index report and receives a Stellar Expert-verifiable wallet-submitted attestation that can be referenced in an audit workflow.
- Independent donor audit: a donor organization or third-party auditor checks a claimed heat-reduction figure against its on-chain hash and timestamp without needing INIT.AI database access.
- Tamper-evidence: any later edit to a previously reported figure produces a different digest under the published canonicalization rules, while the original attestation record remains unchanged on-chain. Corrected reports are new attestations rather than overwrites; where a revision relationship is needed, it is represented by a reference to the prior attestation ID or digest. The original record is never modified, deleted, or replaced.

3.6 Sprint Constraints & Targets

- Open Source: MIT-licensed across the Soroban contract, backend, and dashboard integration code.
- Testnet: all Instaward work is built and validated on Stellar Testnet for this sprint - no mainnet funds are at risk.
- Adoption Target: ≥ 3 live Testnet attestations across ≥ 2 unique tester wallets, with successful confirmation and contract-invocation verification.
- Why Instawards: a bounded 30-day execution sprint that adds a verification layer to an existing platform - not an open-ended platform rebuild.

3.7 How It Works

LGU officer or tester submits a spatial heat report (GeoJSON) via the INIT.AI dashboard

↓

Instawards Statement of Work (SOW)

30-Day Scoped Engagement

Backend validates the GeoJSON, applies the published canonicalization rules, computes the SHA-256 digest, prepares an unsigned `log_heat_index` transaction, and runs transaction simulation before signature is requested.

↓

User reviews the "Verify on Stellar" modal, including the payload summary, digest, non-sensitive metadata, and signing wallet, then signs in their own wallet via `StellarWalletsKit` (Freighter); keys never leave the wallet.

↓

Signed XDR is relayed to Stellar Testnet → `SpatialAttestationRegistry.log_heat_index` records the digest and minimal metadata as a write-once attestation record; raw GeoJSON is not stored on-chain.

↓

The dashboard progresses through explicit states—local preparation, signature requested, transaction submitted, transaction confirmed, and contract invocation verified—and only shows a successful verification state after Stellar Testnet/RPC confirmation.

↓

(Any time after) An auditor independently verifies the digest, signing wallet, timestamp, and transaction/contract invocation on Stellar Expert, without INIT.AI database access.

3.8 Architecture & Trust Model

INIT.AI's verification layer runs in three parts. The existing React + TypeScript + Vite dashboard is extended with `StellarWalletsKit` so all on-chain signing happens client-side and keys never leave the user's wallet; a Python + FastAPI backend validates and canonically serializes GeoJSON heat-report payloads, computes SHA-256 digests, stores relevant off-chain metadata in PostgreSQL/PostGIS on Neon, prepares unsigned XDR, and relays signed XDR to Stellar Testnet RPC; and the `SpatialAttestationRegistry` Soroban contract stores each attestation as a write-once record containing the digest and minimal non-sensitive metadata.

Known trust boundary: the backend can construct but cannot forge a wallet signature. The backend prepares the unsigned XDR from the submitted payload, while every attestation still requires the submitting user's own wallet signature before it reaches the contract. The backend is never a signer and never custodies a private key. However, the backend controls the payload

Instawards Statement of Work (SOW)

30-Day Scoped Engagement

presented for hashing, so a compromised backend could prepare a digest for content different from what the user believes they are attesting to. This sprint keeps that limitation visible in the UI and documentation; fully backend-independent client-side hash recomputation is a roadmap item.

Duplicate definition: for this proof of concept, a duplicate is the same digest within the same defined attestation context (including the LGU reference and reporting-period reference when those fields are present). A legitimate re-attestation outside that context is not treated as a duplicate.

The contract deliberately bounds what any submission can do: each attestation is write-once and scoped to a digest, non-sensitive LGU reference, timestamp, signing wallet, and optional revision reference. An exact duplicate means the same digest within the same defined attestation context (LGU reference and reporting-period reference, where those fields are used). Such a duplicate is rejected rather than silently overwritten. Corrected reports create a new attestation that references the prior attestation ID or digest; earlier records remain unchanged and are never modified, deleted, or replaced. The wallet signature proves control of the signing wallet, not the identity of the signer, LGU authorization, or official certification.

Canonicalization specification (first-class deliverable) and limitations. Before hashing, the versioned canonicalization specification uses RFC 8785-style canonical JSON serialization with lexicographically sorted object keys; UTF-8 encoding; Unicode NFC normalization for string values; coordinates rounded/truncated to 6 decimal places; timestamps normalized to ISO 8601 UTC with seconds and a trailing Z; explicit handling of optional/null fields (null is serialized when the field is present, while omitted optional fields remain omitted); and WGS 84 / EPSG:4326 as the required CRS for attested GeoJSON. The canonical byte sequence is then SHA-256 hashed. The specification will be versioned, documented in plain language, and accompanied by deterministic test vectors so identical reports produce identical digests. The raw GeoJSON, detailed coordinates, imagery, and sensitive LGU operational data remain off-chain. For this sprint, wallet authorization is based on possession of the signing wallet; an authorized-LGU wallet registry, multi-party approval, dispute/correction governance, and oracle-based validation are future roadmap items.

4. Scope of Work (30-Day Deliverables)

Important guidance: This scope must be achievable within **30 calendar days**. If the work feels larger, it should be reduced or split into more achievable phases.

Instawards Statement of Work (SOW)
30-Day Scoped Engagement

INIT.AI Deployment link: init-ai-ebon.vercel.appinit-ai-ebon.vercel.app

4.1 In-Scope Deliverables

Deliverable	Description (What will be built or produced?)	Why this matters
Deliverable 1	SpatialAttestationRegistry Soroban Smart Contract (Testnet): Write, compile, and deploy a Rust-based Soroban contract (SpatialAttestationRegistry.log_heat_index) on Stellar Testnet to record write-once SHA-256 digests and minimal, non-sensitive metadata such as a opaque report ID, digest, version identifier, coarse reporting-period reference where needed, signing wallet, and optional revision reference. The contract rejects exact duplicate attestations and never overwrites prior records.	Establishes the core tamper-evident environmental attestation primitive natively on Stellar while keeping raw GeoJSON off-chain.
Deliverable 2	Dashboard Wallet Integration & Verification UI: Integrate StellarWalletsKit (Freighter extension support) into the INIT.AI React + TypeScript + Vite + Tailwind CSS dashboard. Build the "Verify on Stellar" flow with explicit preparation, signature-requested, submitted, confirmed, and contract-invocation-verified states, plus understandable wallet/RPC/error messages.	Enables non-technical LGU administrators and auditors to sign with their own wallet and understand exactly when an attestation is actually confirmed.
Deliverable 3	Automated Data Hashing Pipeline & Soroban RPC Bridge: Develop Python + FastAPI backend REST API endpoints that validate GeoJSON, apply and document the canonicalization rules, generate SHA-256 digests, persist relevant off-	Bridges INIT.AI's spatial AI processing engine with Stellar while making hashing, simulation, submission, and

Instawards Statement of Work (SOW)
30-Day Scoped Engagement

	chain metadata in PostgreSQL/PostGIS on Neon, prepare unsigned XDR, simulate transactions, relay signed XDR, and monitor Testnet confirmation/contract invocation.	failure states deterministic and reviewable.
Deliverable 4	Ecosystem Validation & Open-Source Package: Release all codebase repositories on public GitHub under the MIT License. Deliver a 3-5 minute technical demo video showing both a successful attestation and at least one expected failure case, an open-source LGU integration guide including the canonicalization specification and trust-boundary disclosure, and execute ≥ 3 live Testnet attestations across ≥ 2 unique tester wallets.	Provides public code auditability and verifiable on-chain proof that the end-to-end flow works across multiple wallets and more than one attestation.
List anything that might be assumed but is not included in this Instaward scope List anything that might be assumed but is not included in this Instaward scope.		
- Mainnet deployment or real mainnet XLM/USDC funds. Live fiat PHP banking or local anchor liquidity integrations. Marketing, promotional events, paid advertising, or community growth campaigns. Paid software subscriptions, paid RPC endpoints, or cloud hosting purchases. Token issuance (\$INIT token) or complex on-chain DAO governance frameworks. Authorized-LGU wallet registry, officer identity verification, multi-party approval, or governance-based role assignment. On-chain dispute resolution or production certification of heat-model accuracy. Storing raw GeoJSON, sensitive coordinates, or other sensitive spatial data on-chain. - Live fiat PHP banking or local anchor liquidity integrations. - Marketing, promotional events, paid advertising, or community growth campaigns. - Paid software subscriptions, paid RPC endpoints, or cloud hosting purchases. - Token issuance (\$INIT token) or complex on-chain DAO governance frameworks.		

Instawards Statement of Work (SOW)
30-Day Scoped Engagement

4.2 Deliverable-Aligned Budget Request

Requested Budget Amount	Rationale for Budget Request
\$4,800 USD	100% Engineering & QA Build Allocation: 160 total hours at a single flat rate of \$30/hour over 30 days. \$0 non-build spend (no software subscriptions, marketing, hosting, or tools requested). • Deliverable 1 (Soroban Contract): \$960 (32 hrs @ \$30/hr) • Deliverable 2 (Wallet UX & UI Modal): \$960 (32 hrs @ \$30/hr) • Deliverable 3 (Data Hashing Pipeline): \$840 (28 hrs @ \$30/hr) • Deliverable 4 (Validation Package & Docs): \$840 (28 hrs @ \$30/hr) • Quality Assurance & Testing: \$1,200 (40 hrs @ \$30/hr)

Instawards Statement of Work (SOW)
30-Day Scoped Engagement

--	--

5. 30-Day Execution Plan & Timeline

5.1 Weekly Breakdown

Week	Planned Work	Expected Output
Week 1	Write Soroban Rust contract code; compile to WASM; deploy SpatialAttestationRegistry to Stellar Testnet; implement write-once duplicate handling and revision references; begin contract-level tests.	Active Testnet Contract ID, public Rust source repository, duplicate/revision tests.
Week 2	Develop the SHA-256 GeoJSON hashing pipeline and publish the exact canonicalization specification; build REST endpoints for unsigned XDR preparation; add transaction simulation and structured RPC error handling.	Functional backend API, canonicalization test vectors, simulation results, and RPC error-state handling.
Week 3	Integrate StellarWalletsKit into the existing React + TypeScript + Vite frontend; build the "Verify on Stellar" flow, wallet signing, and explicit preparation → signature → submission → confirmation → contract-verification states.	Live dApp frontend signing transactions via Freightier with understandable rejection, missing-wallet, timeout, and confirmation states.
Week 4	End-to-end QA; test insufficient XLM, rejected signatures, expired time bounds, RPC failures, simulation failures, contract errors, and duplicate attestations;	Complete validation package, public repositories, documented error handling, canonicalization specification, and Stellar Expert receipts.

Instawards Statement of Work (SOW)
30-Day Scoped Engagement

	execute ≥ 3 confirmed on-chain attestations across ≥ 2 wallets; publish repos, LGU guide, and demo video.	
--	---	--

6. Evidence of Completion (Required)

Important guidance: Evidence should be clear, verifiable, and easy to review by the Ambassador Chapter Lead **with minimal technical expertise**.

Single Evidence Index: The final GitHub README or project page will provide one reviewer-facing table containing the Contract ID, live dApp URL, demo video, API documentation, canonicalization specification, transaction hashes, wallet addresses, and Stellar Expert links.

6.1 Planned Evidence to Be Submitted

Deliverable	Evidence Type (link, repo, demo, screenshot, doc, tx hash, etc.)	Description
Deliverable 1	Contract Address & Repo Link	Public GitHub link to Soroban Rust contract code (MIT License) + live deployed Testnet Contract ID on Stellar Expert, including evidence of duplicate rejection and immutable revision handling.
Deliverable 2	Live dApp URL & Screen Recording	Working frontend deployment link + screen recording showing Freightier connection, payload/digest review, signature request, submission, confirmation, and contract-invocation verification, plus at least one

Instawards Statement of Work (SOW)
30-Day Scoped Engagement

		expected failure case (preferably duplicate-attestation rejection or user signature rejection).
Deliverable 3	API, Canonicalization Spec & Transaction Evidence	API documentation, published canonicalization rules/test vectors, simulation results, and confirmed Testnet transaction hashes on Stellar Expert.
Deliverable 4	Public GitHub Repos & On-Chain Hashes	https://github.com/cseanx/INIT.AI https://github.com/cseanx/INIT.AI Confirmed Testnet receipts across ≥ 2 unique wallet addresses, plus the LGU integration guide and explicit trust-boundary disclosure. The final README/project page will contain one evidence index table covering Contract ID, live dApp URL, demo video, API documentation, canonicalization specification, transaction hashes, wallet addresses, and Stellar Expert links. On-Chain Testnet Receipts: ≥ 3 verified transaction hashes across ≥ 2 unique wallet addresses on Stellar Expert.

6.2 Evidence Verification Checklist (For Ambassador Use)

For each deliverable, the Ambassador Chapter Lead will assess whether evidence is present and sufficient.

Instawards Statement of Work (SOW)
30-Day Scoped Engagement

Deliverable	Evidence Present	Evidence Partial	Evidence Missing	Comments
Deliverable 1	☐	☐	☐	
Deliverable 2	☐	☐	☐	
Deliverable 3	☐	☐	☐	
Deliverable 4	☐	☐	☐	

6.3 Success Metrics

Category	Metric	Target
Adoption targets	Unique tester wallet addresses used to sign attestations	≥ 2
Adoption targets	Live Testnet attestations recorded	≥ 3 , all confirmed and contract-invocation verified
Technical milestones	SpatialAttestationRegistry contract deployed & live on Testnet	Yes
Technical milestones	StellarWalletsKit (Freighter) wallet signing live in dashboard	Yes
Technical milestones	SHA-256 hashing pipeline + RPC bridge live	Yes
Technical milestones	Public LGU integration guide published	Yes
Technical milestones	3-5 minute demo video published	Yes
Technical milestones	All source code released under MIT License	Yes

Instawards Statement of Work (SOW)
30-Day Scoped Engagement

Technical milestones	Canonicalization specification published with deterministic test vectors	Yes
Technical milestones	Transaction simulation and explicit failure-state handling live	Yes
Security / trust	UI discloses backend trust boundary and does not claim underlying heat data is immutable	Yes

7. Security Considerations

7.1 Risks & Mitigations

Risk	Mitigation
Backend presents a mismatched payload for signing	The "Verify on Stellar" modal displays the payload summary, canonical digest, and non-sensitive metadata before signature; the canonicalization specification is public. Full backend-independent client-side recomputation remains a roadmap item.
Compromised or lost tester wallet key	Keys never leave the user's wallet; the backend never custodies or requests a private key and cannot sign an attestation on the user's behalf.
Duplicate or replayed attestation	The contract rejects an exact duplicate digest within the same defined attestation context rather than overwriting it. Corrected reports use a new digest and may reference the prior digest; prior records remain unchanged.
Contract downtime / RPC outage / simulation failure	Transactions are simulated before signature where supported; failures surface explicit retry/error states. The dashboard distinguishes preparation, signature request,

Instawards Statement of Work (SOW)
30-Day Scoped Engagement

	submission, confirmation, and contract-invocation verification.
Malformed or invalid GeoJSON payload	The backend validates payload structure, CRS, coordinate precision, timestamp format, and canonicalization requirements before hashing and rejects invalid submissions before transaction preparation.
Wallet is not proof of LGU officer authorization	For this sprint, wallet signing proves control of the signing wallet only; it does not establish signer identity, LGU authorization, or official certification. No LGU identity registry or officer-role verification is implemented; this limitation is disclosed in the UI/docs and is a future roadmap item.
Sensitive spatial data or identifiers exposed on-chain	Only the digest and reviewed minimal non-sensitive metadata are written on-chain. Raw GeoJSON, detailed coordinates, imagery, and sensitive LGU operational data remain off-chain; LGU references used on-chain must be pseudonymous/non-sensitive.
Expired transaction time bounds / rejected signature / insufficient XLM	The UI surfaces each condition distinctly, provides a retry path where safe, and never labels a transaction successful until Testnet confirmation and contract-invocation verification are received.

7.2 Security Controls

- Non-Custodial Key Isolation: Private keys remain exclusively client-side in StellarWalletsKit (Freighter).
- The FastAPI backend prepares unsigned transaction XDRs, runs validation/simulation, and relays signed XDRs; it never stores, requests, or handles private keys.
- Write-Once Soroban State Enforcement: The SpatialAttestationRegistry enforces append-only record creation, rejects exact duplicate attestations, and never overwrites historical records.
- Deterministic Canonicalization & Schema Validation: Payloads undergo strict pre-validation using the published rules—RFC 8785-style canonical JSON serialization, lexicographical object-key ordering, UTF-8 encoding, Unicode NFC normalization, coordinates rounded/truncated to 6 decimals, timestamps normalized to ISO 8601 UTC seconds with Z, explicit omitted-vs-null handling, and WGS 84 / EPSG:4326 CRS—before SHA-256 hashing.

Instawards Statement of Work (SOW)

30-Day Scoped Engagement

- Testnet Network Enclosure: Smart contract invocation endpoints and RPC listeners are hardcoded to Stellar Testnet, guaranteeing zero exposure of mainnet accounts or real financial assets during the sprint.
- Backend Trust Visibility & State Accuracy: The UI displays the payload summary and computed SHA-256 hash before requesting signature, explicitly discloses the backend trust boundary, and separates local preparation, signature requested, submitted, confirmed, and contract-invocation verified states. Success is shown only after Testnet confirmation.

8. Final Success Criteria

This Instaward is complete when all the following are true and verifiable:

- Deliverables completed - all four deliverables (Soroban attestation contract, wallet integration & verification UI, hashing/RPC bridge, and ecosystem validation package) are shipped with the canonicalization specification and trust-boundary disclosure.
- Validation achieved - ≥ 3 live Testnet attestations across ≥ 2 unique tester wallets, each confirmed and contract-invocation verified on Stellar Expert, plus successful duplicate-rejection and immutable revision-path tests, with at least one expected failure case demonstrated in the final demo.
- Publicly accessible solution - the live dashboard's wallet-signing flow, canonicalization specification, and public LGU integration guide are reachable by anyone.
- Measurable Stellar ecosystem impact - the sprint produces real Testnet Soroban contract invocations and auditable attestation digests while demonstrating that raw spatial heat data remains off-chain and that the UI does not claim confirmation before RPC verification.

9. Next-Step Alignment

9.1 Anticipated Next Step After Completion

After this Instaward, the most likely next step is:

- **[X] Apply to SCF Build Award**

Instawards Statement of Work (SOW)
30-Day Scoped Engagement

- ☐ Continue development independently
- ☐ Apply for a follow-on Instaward (if eligible)
- ☐ Seek other ecosystem support
- ☐ Other:

10. Instawards Constraints Acknowledgement

By submitting this SOW, the Builder acknowledges:

- ☒ This scope will be completed within **30 days or less**.
- ☒ Instawards support execution, not open-ended exploration.
- ☒ A project may receive **no more than two follow-on Instawards**.
- ☒ Each Instaward is capped at **\$5,000**.
- ☒ Total Instawards funding may not exceed **\$15,000**.

11. Submission Confirmation

Once finalized, this Statement of Work will be submitted by the Ambassador Chapter Lead via the Instawards Airtable submission form for review and approval.

12. Basis of Estimate

12.1 Basis of Estimate

Parameter	Value
Sprint duration	30 days (4 working weeks)
Hourly rate (all work)	\$30 / hour - one flat rate for every hour in this budget; no blended, premium, or role-differentiated rates

Instawards Statement of Work (SOW)
30-Day Scoped Engagement

Development (Soroban + full-stack)	\$1,200 - 40 hours of contract, wallet/UI, hashing/canonicalization, RPC/error-state, privacy/trust-boundary, and end-to-end Testnet validation testing
QA & testing	\$1,200 - 40 hours of contract, wallet/UI, pipeline, and end-to-end testnet validation testing
Total	\$4,800 USD - 160 hours × \$30/hour

12.2 Cost by Deliverable (Development - \$3600, 120 hours at \$30/hours)

Deliverable 1 - SpatialAttestationRegistry Soroban Contract - \$960 (32 hrs)

Work Package	Hrs	Cost
Rust contract design & write (log_heat_index function, digest/minimal-metadata storage schema, duplicate/revision behavior)	10	\$300
Local build & unit tests (Soroban SDK)	6	\$180
Testnet deployment & contract ID verification	4	\$120
Access-control / input-validation hardening and duplicate/revision tests	6	\$180
Integration test: submit attestation from an external wallet, capture tx hash	6	\$180
Subtotal	32	\$960

Deliverable 2 - Dashboard Wallet Integration & Verification UI - \$960 (32 hrs)

Work Package	Hrs	Cost
StellarWalletsKit integration (Freighter connect / sign)	8	\$240
"Verify on Stellar" UI modal (payload/digest review and explicit preparation/signature/submission/confirmation/contract-verification states)	8	\$240
Signing/submission UX states (missing wallet, rejected signature, pending, submitted, confirmed, contract verified, error, tx link)	8	\$240
Signing/submission UX states (pending, confirmed, error, tx link)	5	\$150
Cross-browser / wallet QA of the signing flow	3	\$90

Instawards Statement of Work (SOW)
30-Day Scoped Engagement

Subtotal	32	\$960
----------	----	-------

Deliverable 3 - Automated Data Hashing Pipeline & Soroban RPC Bridge - \$840 (28 hrs)

Work Package	Hrs	Cost
SHA-256 digest generation from canonically serialized spatial heat-report payloads	6	\$180
RPC bridge: transaction simulation, signed-XDR relay, read/write requests to Stellar Testnet	5	\$150
Retry / error-handling for RPC failures, time bounds, insufficient XLM, simulation failures, contract errors, and duplicate attestations	8	\$240
Retry / error-handling & network fallback logic	5	\$150
Pipeline integration test (payload → hash → on-chain write)	4	\$120
Subtotal	28	\$840

Deliverable 4 - Ecosystem Validation & Open-Source Package (development portion) - \$840 (28 hrs)

Work Package	Hrs	Cost
LGU integration guide - canonicalization specification, sample payloads, worked example, and trust-boundary disclosure	8	\$240
Tester-wallet onboarding support (≥ 2 unique wallets, ≥ 3 confirmed attestations, duplicate/revision test cases)	6	\$180
Tester-wallet onboarding support (≥ 2 unique wallets, ≥ 3 attestations)	6	\$180
Demo-video capture + collation of on-chain receipts	4	\$120
MIT license / repo cleanup & publication	4	\$120
Subtotal	28	\$840

QA & testing - 40 hrs / \$1,200 at the same \$30/hour - is itemized separately in A.3.

Development total (D1-D4): 120 hrs / \$3,600. Combined total: 160 hrs / \$4,800.

12.3 Cost by Category (Reconciles to SOW 4.2)

Category (SOW 4.2 line)	Components	Amount
Development (Soroban + full-stack)	D1 \$960 + D2 \$960 + D3 \$840 + D4 \$840 (see A.2) - 120 hrs at \$30/hour	\$3,600

Instawards Statement of Work (SOW)
30-Day Scoped Engagement

QA & testing	Contract-level unit & testnet QA (10 hrs); wallet/UI cross-browser and state QA (10 hrs); hashing/canonicalization/API QA (10 hrs); end-to-end testnet validation, error-state testing, and on-chain receipt verification (10 hrs)	\$1,200
Total	160 hours at \$30/hour	\$4,800

12.4 Cost by Week (Timeline-aligned)

Week (SOW \$5.1)	Primary Work	Development	QA & Testing	Week Total
Week 1 - Soroban contract	Deliverable 1	\$960 (32 hrs)	\$300 (10 hrs)	\$1,260 (42 hrs)
Week 2 - Hashing pipeline & RPC bridge	Deliverable 3	\$840 (28 hrs)	\$300 (10 hrs)	\$1,140 (38 hrs)
Week 3 - Wallet integration & UI	Deliverable 2	\$960 (32 hrs)	\$300 (10 hrs)	\$1,260 (42 hrs)
Week 4 - Validation package	Deliverable 4	\$840 (28 hrs)	\$300 (10 hrs)	\$1,140 (38 hrs)
Total	160 hours at \$30/hour	\$3,600 (120 hrs)	\$1,200 (40 hrs)	\$4,800 (160 hrs)